

Taxonomy construction and refinement

□ Taxonomy

- Organizing important concepts into semantically rich structures
- Playing an essential role at organizing massive unstructured text data into relatively organized structures

□ Suitable taxonomy properties

- multi-faceted
- corpus- and application-dependent

□ Taxonomy generation and expansion

- Weakly or distantly supervised methods
 - Set expansion
 - Embedding-based hierarchical clustering
 - Adding new emerging terms or deleting old, obsolete ones

Weakly supervised text classification

- ❑ Text: massive, diverse and in multiple granularities
- ❑ Weakly supervised text classification
 - ❑ Idea #1:
 - ❑ Generate pseudo documents by Embedding methods
 - ❑ Train neural networks based on the pseudo documents and unlabeled text data
 - ❑ Idea #2:
 - ❑ Generate class-distinctive keywords or phrases by category-guided embedding
 - ❑ Idea #3:
 - ❑ Pre-trained language model to generate class-distinctive keywords

Fine-grained information extraction

- ❑ Fine-grained entity recognition
 - ❑ Extract the concrete role that an entity plays in a particular context
- ❑ Identification of the context of an entity
 - ❑ Generate entity mention candidates and relation phrases
 - ❑ Find the most appropriate fine-grained types for the entities to be examined
- ❑ Methods
 - ❑ Distant supervision based methods
 - ❑ Embedding based methods
 - ❑ Taxonomy-guided supervision and pre-trained language model

Knowledge graph/information network construction

- ❑ Knowledge graphs
 - ❑ A set of entities
 - ❑ A set of (possibly labeled) edges linking among entities
- ❑ Construction of knowledge graph
 - ❑ Use distant or weak supervision
 - ❑ Taxonomy-guided text classification
 - ❑ Problems of building knowledge graph
 - ❑ An entity can be associated with different attributes/value
 - ❑ Easy to cause confusion if different associations are merged into a single knowledge graph
 - ❑ Solutions
 - ❑ Construct local knowledge graphs
 - ❑ Use such local knowledge graphs under similar conditions